Carson Hedrich

carsonhedrich.github.io • carsonhedrich@gmail.com • (239) 989-8301

EDUCATION

Georgia Institute of Technology

Graduated December 2025

B.S. Computer Science, *summa cum laude* - Cybersecurity Concentration | GPA: 3.87

Relevant Coursework: Machine Learning Security, Systems & Networks, Network Security, Critical Infrastructures, Malware Reverse Engineering, Information Security

SKILLS

Malware Analysis:

IDA Pro, Ghidra, OllyDbg - static/dynamic analysis, sandbox unpacking, binary extraction, assembly-level reverse engineering

Penetration Testing & Offensive Security:

Nmap, Metasploit, ffuf, Gobuster, Hydra - SSRF, CSRF, XSS, SQL injection, reverse shells, privilege escalation

Digital Forensics & Defense:

Autopsy, Splunk (SIEM) - evidence collection, log analysis, alert triage, threat detection, network packet analysis

Languages & Platforms:

Java, C++, C, Python, Assembly, Dart, CSS, JavaScript - Linux, Windows

EXPERIENCE

Google - Student Training in Engineering Program (STEP) Intern | New York, NY

Summer 2023

- Extended the back-end and front-end of an advertiser campaign insights feature that surfaces ad performance trend data to millions of Google Ads users.
- Wrote 10+ unit and end-to-end tests; contributed multiple Java and Dart classes across both layers of the feature, covering back-end data processing and front-end rendering.
- Participated in the full software development lifecycle: authored design documents, completed design reviews, implemented the feature, and produced full documentation for continued maintenance.
- Collaborated with one fellow intern to scope and deliver the feature end-to-end, using Java, Dart, CSS, and Mockito.

PROJECTS

Malware Reverse Engineering: "Harulf" Virus Analysis

Spring 2025

- Performed static and dynamic analysis on a packed, polymorphic Windows PE virus to determine its infection mechanism and payload behavior.
- Applied OllyDbg sandboxing and debugging techniques to safely unpack the virus and extract the unencrypted binary, then conducted assembly-level analysis in IDA Pro.
- Identified and documented the virus's polymorphic evasion capabilities, producing findings applicable to improving antivirus threat definitions.

Digital Forensics: Cybercrime Investigation Semester Project

Fall 2024

- Led a simulated cybercrime investigation using Autopsy, recovering evidence from steganography-concealed messages, obfuscated files, suspicious web history, and network packet traces.
- Exploited vulnerabilities in simulated suspect websites to extract additional evidence and establish proof of illegal activity.

PROFESSIONAL DEVELOPMENT

TryHackMe — Online Cybersecurity Training Platform

September 2025 – Present

- Ranked in the top 20% of platform users; actively progressing through the Jr. Penetration Tester and Security Engineer learning paths.
- Building hands-on skills in offensive and defensive security through labs that simulate real-world attack scenarios and incident response.
- Practicing ethical penetration testing using Nmap, Hydra, Gobuster, and Metasploit to discover and exploit vulnerabilities in controlled environments.